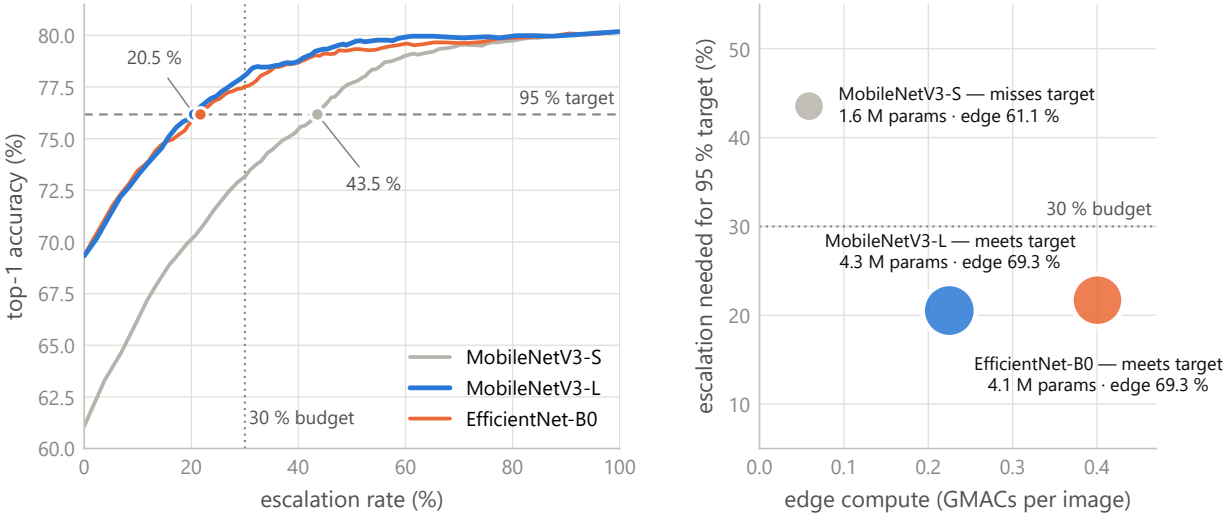
(a) accuracy vs. escalation, margin router
top-1 accuracy (%)
escalation rate (%)
80.0
77.5
75.0
72.5
70.0
67.5
65.0
62.5
60.0
0
20
40
60
80
100
20.5 %
95 % target
43.5 %
30 % budget
MobileNetV3-S
MobileNetV3-L
EfficientNet-B0
(b) cost of the edge vs. escalation it saves
escalation needed for 95 % target (%)
edge compute (GMACs per image)
50
40
30
20
10
0.0
0.1
0.2
0.3
0.4
MobileNetV3-S — misses target
1.6 M params · edge 61.1 %
30 % budget
MobileNetV3-L — meets target
4.3 M params · edge 69.3 %
EfficientNet-B0 — meets target
4.1 M params · edge 69.3 %